

TALAL KHAN

Senior Backend Software Engineer

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PROFESSIONAL SUMMARY

Results-driven Senior Backend Software Engineer with 3+ years of experience architecting, developing, and optimizing high-performance backend services and cloud-native applications. Demonstrated success achieving 99.9% uptime and reducing API response times by 40% through strategic optimizations, Redis caching, and async/await refactoring. Expert in Node.js, TypeScript, microservices architecture, serverless computing (AWS Lambda, EventBridge), and containerized deployments (Docker, AWS ECS/ECR). Proficient in CI/CD pipeline automation (GitHub Actions), observability and monitoring (Datadog, AWS CloudWatch), CDN and edge delivery (AWS CloudFront), multi-cloud infrastructure management (AWS, GCP, DigitalOcean), and event-driven system design. Knowledgeable in OWASP security standards including JWT/OAuth2 authentication, input validation, and rate limiting for building secure, resilient APIs. Proven ability to lead cross-functional engineering teams and deliver scalable, resilient distributed systems within Agile/Scrum frameworks.

CORE COMPETENCIES

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|------------------------------|----------------------------------|------------------------------------|
| • Backend Development | • Cloud Infrastructure (AWS/GCP) | • Agile/Scrum Methodologies |
| • Microservices Architecture | • CI/CD Pipeline Automation | • Performance Tuning |
| • Serverless Computing | • Containerization (Docker/ECS) | • GraphQL API Development |
| • RESTful API Design | • Database Optimization | • Payment Integration |
| • Event-Driven Architecture | • System Design & Scalability | • Web Application Security (OWASP) |

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Node.js, Python, GoLang (Go)

Backend Frameworks: Express.js, Nest.js, FastAPI, Django, Next.js, React.js

Databases: PostgreSQL, MongoDB, MySQL, Redis, SQL

AWS Services: Lambda, EventBridge, ECR, ECS/Fargate, EC2, Elastic Beanstalk, SQS, SNS, S3, API Gateway, MediaConvert, CloudFront, CloudWatch, IAM

Cloud Platforms: Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure, DigitalOcean, Heroku

Observability & Monitoring: Datadog (APM, Logs, Infrastructure Monitoring, RUM), AWS CloudWatch, Real-Time Alerting

DevOps & Tools: Docker, CI/CD, GitHub Actions, SonarQube, Postman, Swagger/OpenAPI, Jest, Turborepo, Monorepo

Architecture & Patterns: Microservices Architecture, Serverless Architecture, Event-Driven Design, RESTful APIs, GraphQL, gRPC, System Design, OOP

Security: OWASP Top 10, JWT, OAuth2, Input Validation, Rate Limiting, XSS Prevention, CSRF Protection, SQL Injection Prevention

Integrations & Payments: Stripe, PayPal, Benefit-Gateway, Meta APIs (Instagram, Messenger, WhatsApp), Cloudinary, Bull-MQ, WebSocket, Socket.io

AI & Developer Tools: Claude Code, ChatGPT/OpenAI, Cursor IDE, GitHub Copilot, Anti-Gravity

PROFESSIONAL EXPERIENCE

Senior Software Engineer & Team Lead

Tafsol Technologies, Karachi, Pakistan

January 2022 – May 2026

- Architected and maintained high-throughput backend services processing production-scale traffic, guaranteeing 99.9% uptime across distributed environments using Node.js, Express.js, and Nest.js
- Reduced average API response times by 40% through strategic query optimization, Redis caching implementation, and async/await refactoring across critical service endpoints
- Implemented OWASP security best practices including JWT/OAuth2 authentication, input sanitization, and rate limiting to safeguard APIs against common vulnerabilities (XSS, CSRF, SQL Injection)
- Engineered serverless event-driven workflows using AWS Lambda and Amazon EventBridge, automating scheduled tasks and reducing operational overhead by 60%
- Containerized microservices with Docker and orchestrated deployments via AWS ECR and Amazon ECS (Fargate), enabling auto-scaling and zero-downtime releases
- Established automated CI/CD pipelines using GitHub Actions integrating ECR image pushes and ECS service updates, cutting deployment time by 50% and eliminating manual errors
- Led cross-functional team of 5+ engineers, delivering features within sprint schedules adhering to Agile/Scrum methodologies and conducting architectural design reviews

Associate Software Engineer

Friendsbit Software Solutions, Karachi, Pakistan

October 2021 – January 2022

- Built and maintained Node.js backend applications, reducing codebase complexity by 20% through modular refactoring while retaining full functionality
- Optimized GraphQL queries and data-fetching patterns, achieving a 30% reduction in page load times and improved client-side performance
- Contributed to architectural design reviews, ensuring adherence to scalability, maintainability, and microservices best practices

Software Engineer

Objectual Systems Ltd, Karachi, Pakistan

August 2020 – September 2021

- Enhanced backend process efficiency through systematic performance tuning and code optimization, reducing core service response times by 25%
- Refactored legacy modules across 3+ products, decreasing bug-report frequency by 30% and improving overall system stability and maintainability
- Collaborated on multiple product teams, delivering features that augmented system functionality and improved user satisfaction metrics

EDUCATION

Bachelor of Science, Computer Science

Federal Urdu University of Arts, Science & Technology, Karachi, Pakistan

2019 – 2024

CERTIFICATIONS

- AWS Cloud Practitioner Essentials – Amazon Web Services (AWS)
- Node.js Course – Mosh Hamedani
- Certified Computer Operator – Saylani Mass IT Training